

Exercise 1.F: From Design to Implementation

How do we implement the fixed Part E architecture as real retrieval code?

Part F implements the same multimedia retrieval architecture twice over Flickr8k.

Classical MDM / classical ML pipeline
explicit descriptors
required k-NN over handcrafted visual features

Modern deep-learning pipeline
CLIP ViT-B/32 learned embeddings
FAISS IndexFlatIP retrieval

Validation note

FAST_DEV runtime validation passed on a 200-image subset.
The full Flickr8k run is explicitly deferred.
No final benchmark conclusion is drawn from these slides.

representation choice

indexing choice

query processing

evaluation design

Why Flickr8k Fits Exercise 1.F

What is the multimodal object?

Multimedia object

one image
five captions
metadata

Structured store

image id, width, height, aspect ratio

Semi-structured store

five caption fields per image
treated as a list of text fields

Media/content DB

JPEG image content

Text query -> retrieve images

Image query -> retrieve visually
similar images

Text + image query -> fused
retrieval

Bridge to Part E: Flickr8k instantiates the fixed Part E architecture using real multimedia objects.

Reference: content/lab/1E/slide-E-draft-ali_merge 2.pdf

Pipeline 1: Classical MDM / Classical ML Pipeline

How can we retrieve images without deep learning?

Pipeline 1 is a classical MDM / classical ML pipeline: explicit representations plus k-NN over handcrafted features. It excludes deep learning, not machine learning.

Text path

concatenate five captions
compute TF-IDF vectors
rank by cosine similarity

Visual path

extract HSV color histogram
extract Canny/edge-density descriptor
classical ML: required k-NN over
handcrafted descriptors

Fusion path

combine text and visual scores
only for real dual text+image queries

Forbidden in Pipeline 1

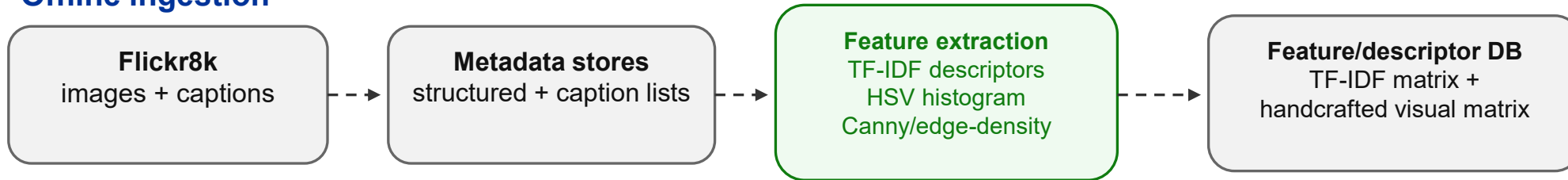
CLIP, SigLIP, BERT, CNN/transformer encoders, neural embeddings/classifiers, depression-detection components
VNN, OCR, NRClex, Empath, XGBoost

The baseline is deliberately interpretable and instance-based for the visual retrieval component.

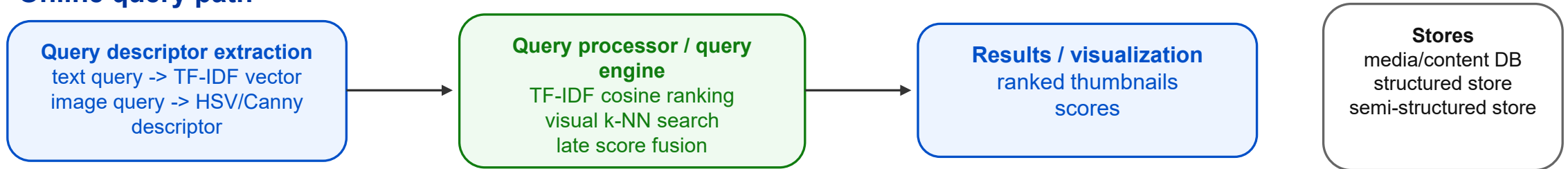
Pipeline 1 Mapped to the Fixed Part E Architecture

Where does the classical ML code sit in the architecture?

Offline ingestion



Online query path

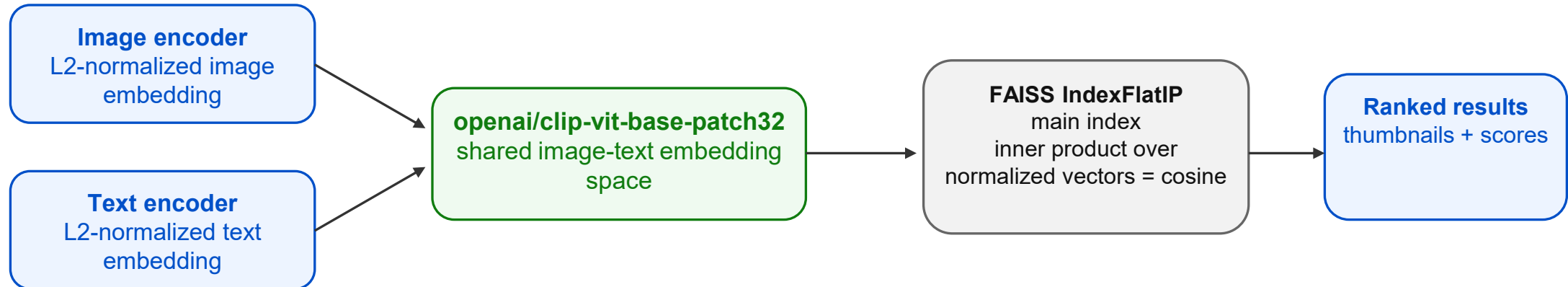


Canny/edge-density is a nested low-level visual step, not a top-level architecture module.

Pipeline 2: Modern Deep-Learning CLIP + FAISS

What changes in the modern deep-learning pipeline?

Pipeline 2 is the modern deep-learning pipeline: CLIP ViT-B/32 learns image/text embeddings; FAISS retrieves vectors.



Optional scalability discussion

IVF/nprobe may be discussed as an approximate-search extension; not the main Flickr8k setting.

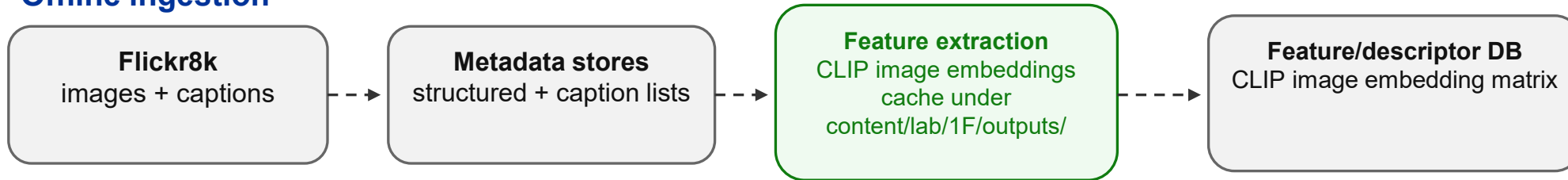
Guardrails

SigLIP2 is background/optional only. ViT patching inside CLIP is not the Part E image-segmentation module.

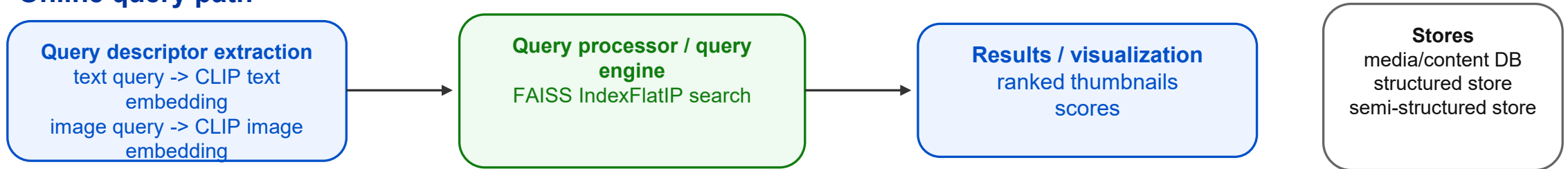
Pipeline 2 Mapped to the Fixed Part E Architecture

How does the modern deep-learning pipeline still follow Part E?

Offline ingestion



Online query path



Key point: The architecture is reused; the representation changes.

What Stayed Constant, What Changed, and How We Evaluate

What makes the comparison fair?

Constant across both pipelines

same dataset: Flickr8k
same query file: evaluation_queries.json
same three evaluation paths
same metrics: Precision@K, Recall@K

Evaluation paths

evaluate_text_only(...)
evaluate_visual_only(...)
evaluate_fused(...)
alpha only in fused retrieval

Evaluation contract

required caption terms define relevance
optional terms do not define binary relevance
TF-IDF similarity is not ground truth
reference image excluded

Important limitation

visual/fused query anchors currently use placeholder reference ids with deterministic fallback.
Replace them with real Flickr8k filenames before treating visual/fused metrics as final.

Takeaway: Expected Trade-offs, Not Final Benchmark Claims

What should students learn?

Classical ML over explicit descriptors

- explicit and inspectable descriptors
- interpretable similarity logic
- direct connection to classic MM indexing
- less semantic generalization by construction

Deep learning over learned multimodal embeddings

- image-text semantic embedding space
- less manual feature engineering
- vector-index-based retrieval
- lower interpretability and model-dependent behavior

No final comparison verdict here

FAST_DEV validates the implementation workflow only. Full Flickr8k benchmarking is deferred.

Part F implements the fixed Part E retrieval architecture twice: explicit representations + classical ML, and CLIP deep-learning encoder + FAISS retrieval.